



DO-IT-YOURSELF: CIRCUIT IDEA

illumination. If a short delay is selected, the LED strip turns on for about 6 seconds after dusk, then turns off for 6 seconds, repeating this cycle throughout the night until dawn.

Thus, the circuit provides three selectable timing intervals, short, medium, and long, enabling automatic switching of the LED strip on/off at dusk with a chosen delay, making it suitable for energy-efficient, unattended lighting applications.

Construction

An actual-size, single-sided PCB layout for the auto dusk-to-dawn LED strip on/off with a timer is shown in Fig. 3, and its component layout is shown in Fig. 4. Alternatively, the circuit can be assembled on a high-quality general-purpose PCB if a custom PCB is not available.

After completing the assembly, calibrate the sensitivity. Power

on the circuit and keep the LDR exposed to normal daylight. Slowly adjust the preset VR2 until the LED strip just turns off. Next, cover the LDR by hand or place the unit in a dim environment to simulate darkness. If the LED strip turns on, the threshold is correctly set. Fine-tune VR2 so that switching occurs exactly at the desired dusk light level. This ensures reliable dusk-to-dawn operation without false triggering.

Next, test the timer section. With the LDR in dark conditions, select different switch S1 positions (C3, C4, C5). Measure the on-duration of the LED strip using a stopwatch. The delay time should vary with the selected capacitor value (larger capacitance yields a longer delay). If required, minor adjustments can be made by slightly varying R8 or R9 values to obtain the desired timing range. Check for stable operation over several cycles to confirm proper

calibration and reliable performance.

Now, mount the assembled PCB inside a suitably sized plastic or ABS cabinet. For user convenience, fix JACK1 (12V DC input jack) and S1 (SP3T time-selection switch) on the front panel of the cabinet. Extend LED1 to the front panel for easy monitoring and label switch S1 for time selection (short/medium/long). After final assembly, connect the LED strip to CON1 with the correct polarity.

After calibration, testing, and final assembly, the unit is ready for installation in corridors, gardens, entry gates, or signboards for automatic dusk-to-dawn lighting control. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/auto-dusk-to-dawn-led-strip-on-off-with-timer>

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